

The role of pseudo-autosomal region reads mapping in sex and age aware glia cell-type deconvolution in scRNA-seq

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Introduction

Cell type annotation:

Marker

- Biologically robust
- Low-resolution
- Slow

Reference transcriptome

- Depend on robustness of reference
- High-resolution
- Fast

Reference: ([Heumos et al, 2023](#); [He et al, 2022](#))

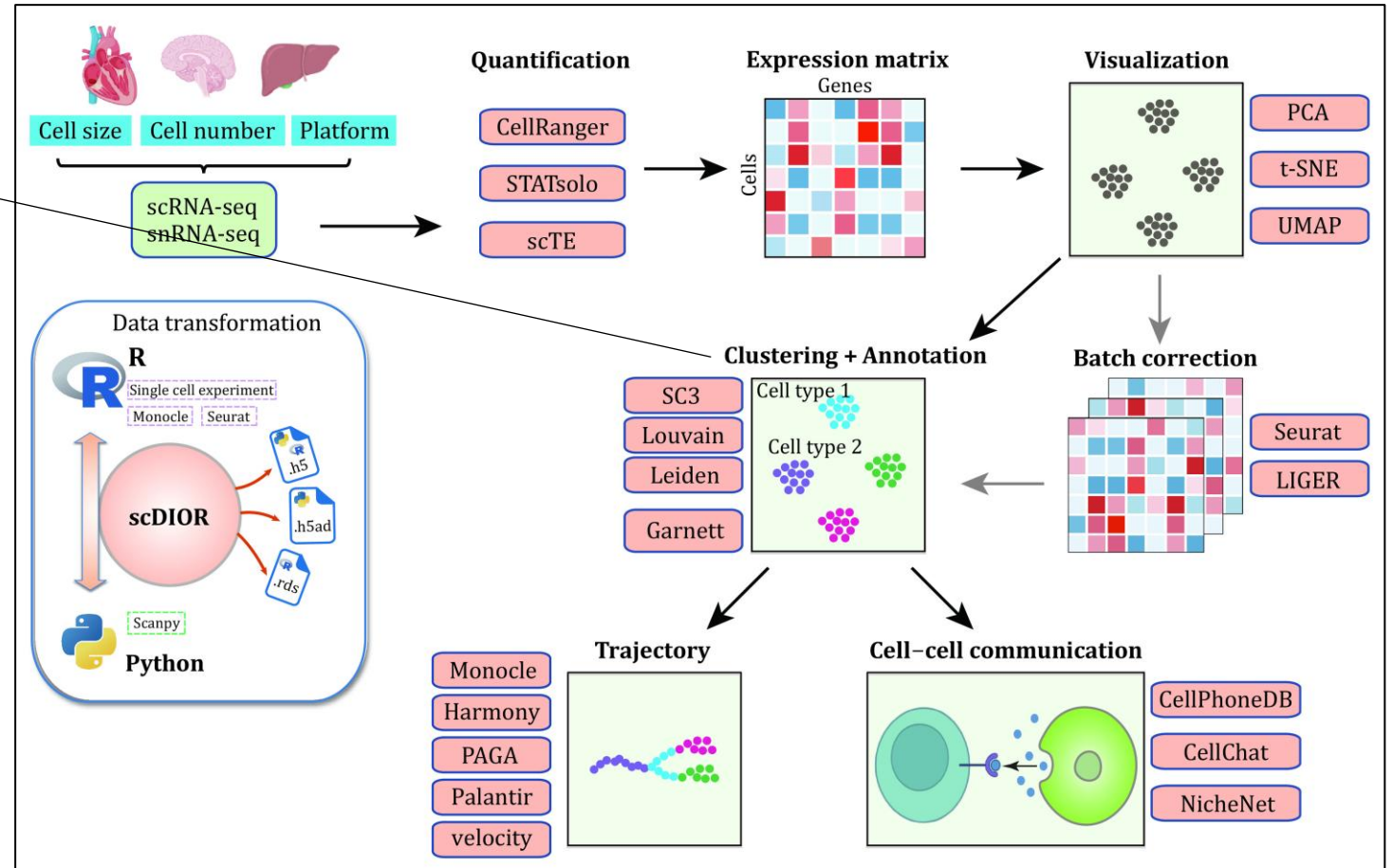


Figure 1: Overview of scRNA-seq data analysis workflow ([He et al, 2022](#)).

Introduction

- Allen Brain Atlas
- Jin et al, 2025
(<https://doi.org/10.1038/s41586-024-08350-8>)
 - 108 mice (44 aged, 64 adult; 51 female, 57 male)
 - ~35% coverage of brain volume
 - ~2 million cells post-quality control

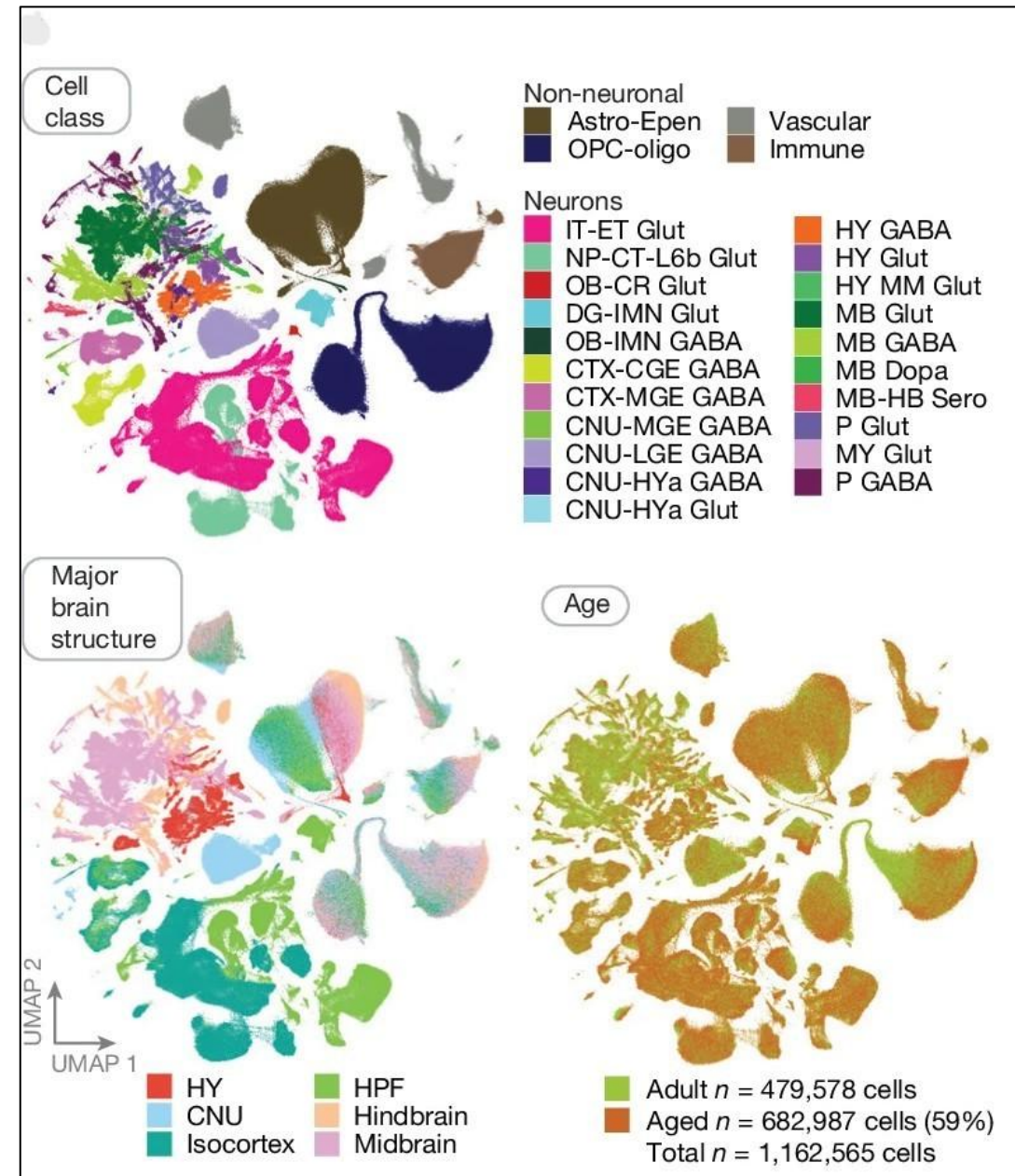


Figure 2: Umap visualization of whole atlas ([Jin et al, 2025](#), figure 1e).

Introduction

- Global clustering → Diluted biology (False negative)
- Over clustering → Over confident “novel” cell type (False positive)

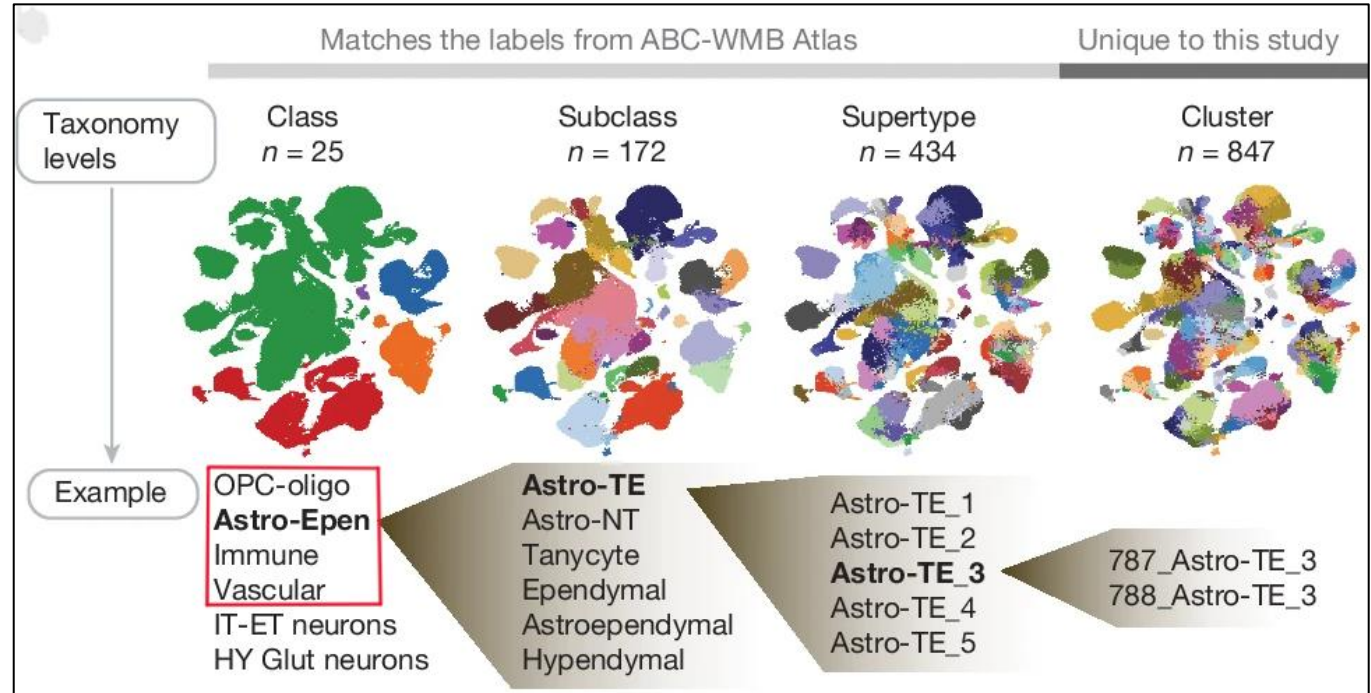


Figure 3: Hierarchical cell type annotation; Red box indicate all non-neuronal class ([Jin et al, 2025](#), figure 1c).

Introduction

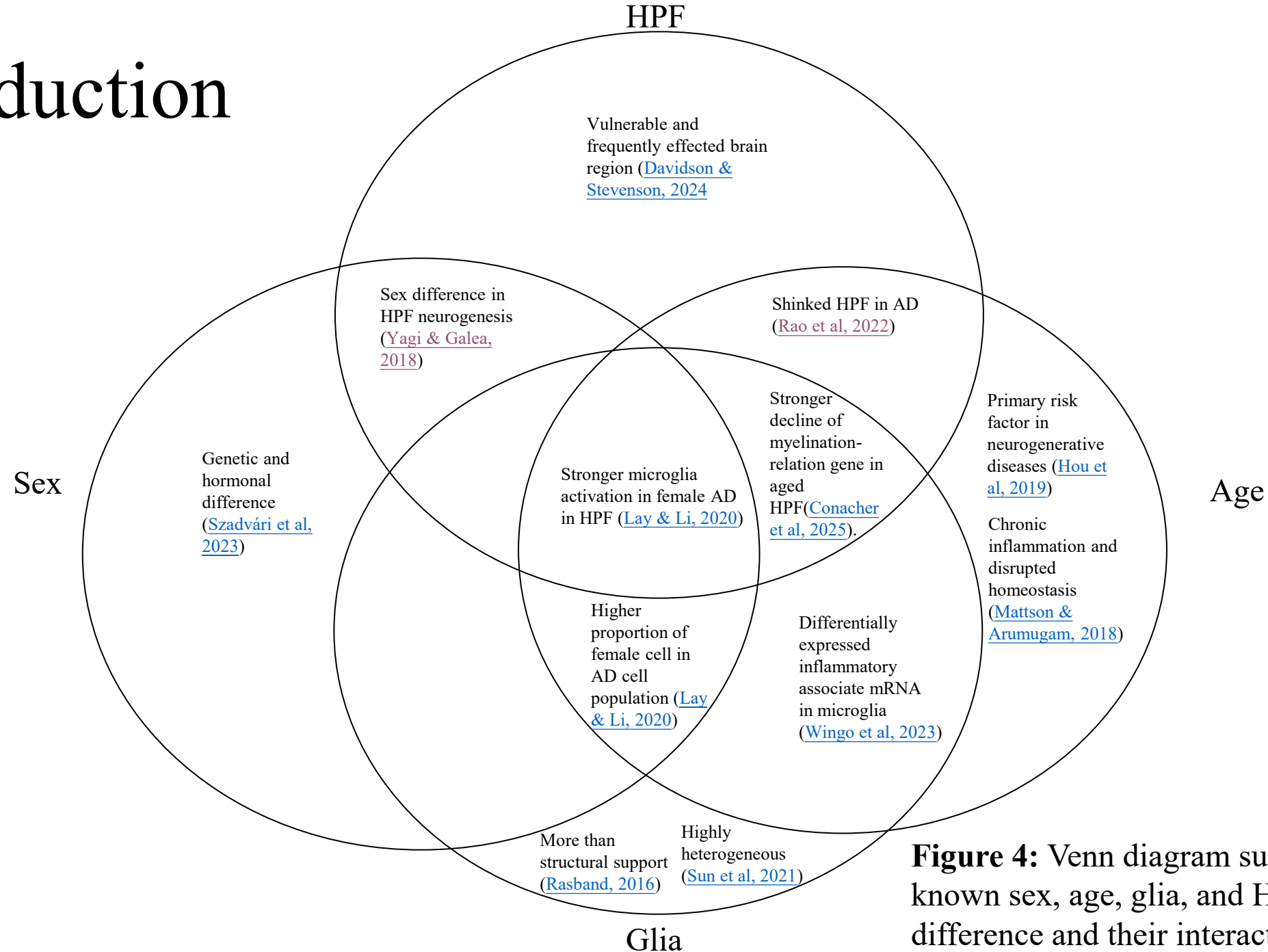


Figure 4: Venn diagram summarize known sex, age, glia, and HPF difference and their interaction.

Method

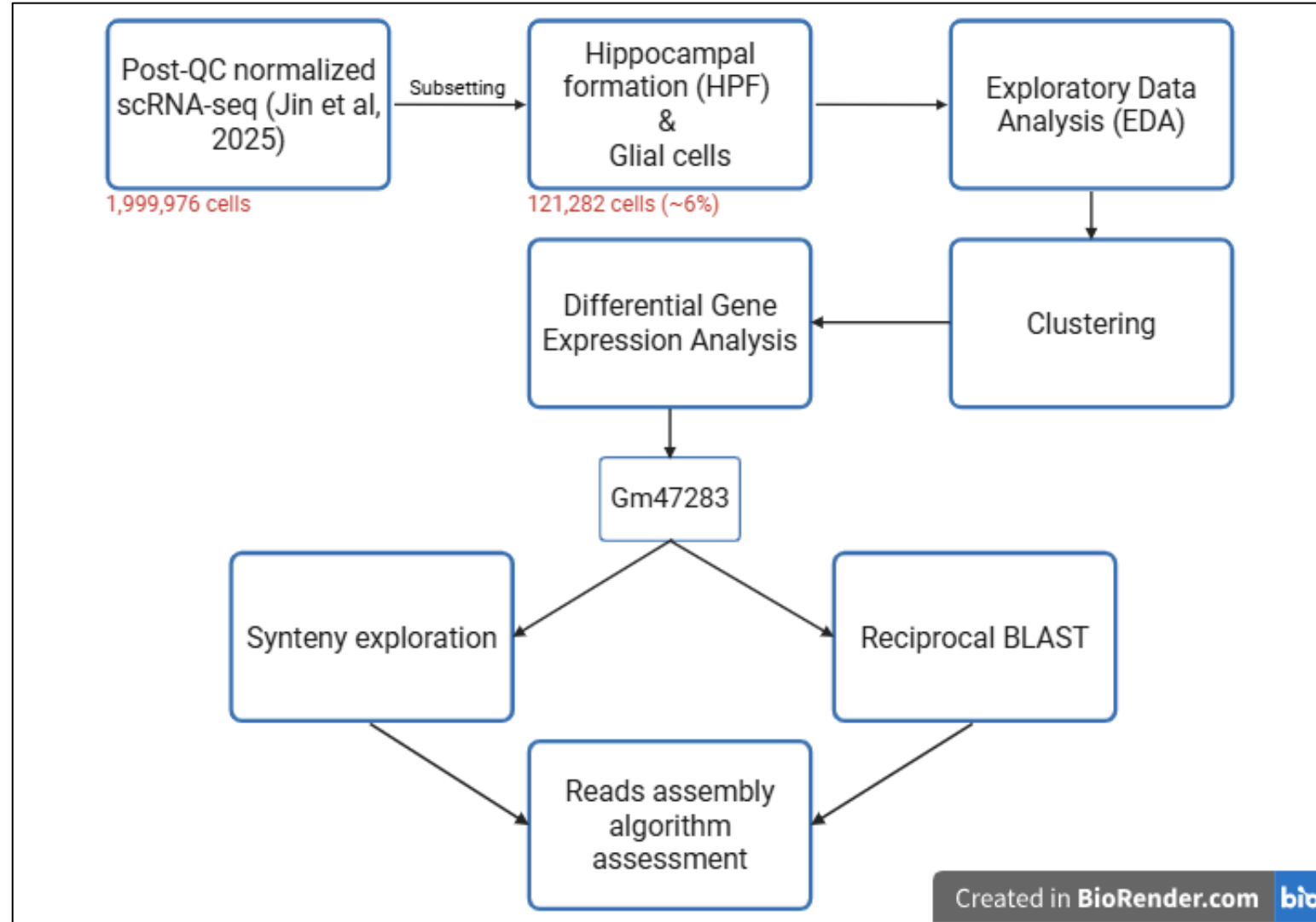


Figure 5: Method schematic; Number of cells retained in first two steps are indicated in red (Created in <https://BioRender.com>).

Result

Exploratory Data Analysis

ENT: Entorhinal cortex
HIP: Hippocampus
HIP - CA: Cornu Ammonis
PAR_POST_PRE_SUB_ProS: Hippocampal-subicular

Subclass:

- 16 subclass
- Astrocyte dominated

Region of interest (roi):

- 4 roi (only 3 roi documented in [Jin et al, 2025](#))
- ENT dominated

Sex:

- Male biased

Age:

- Aged biased

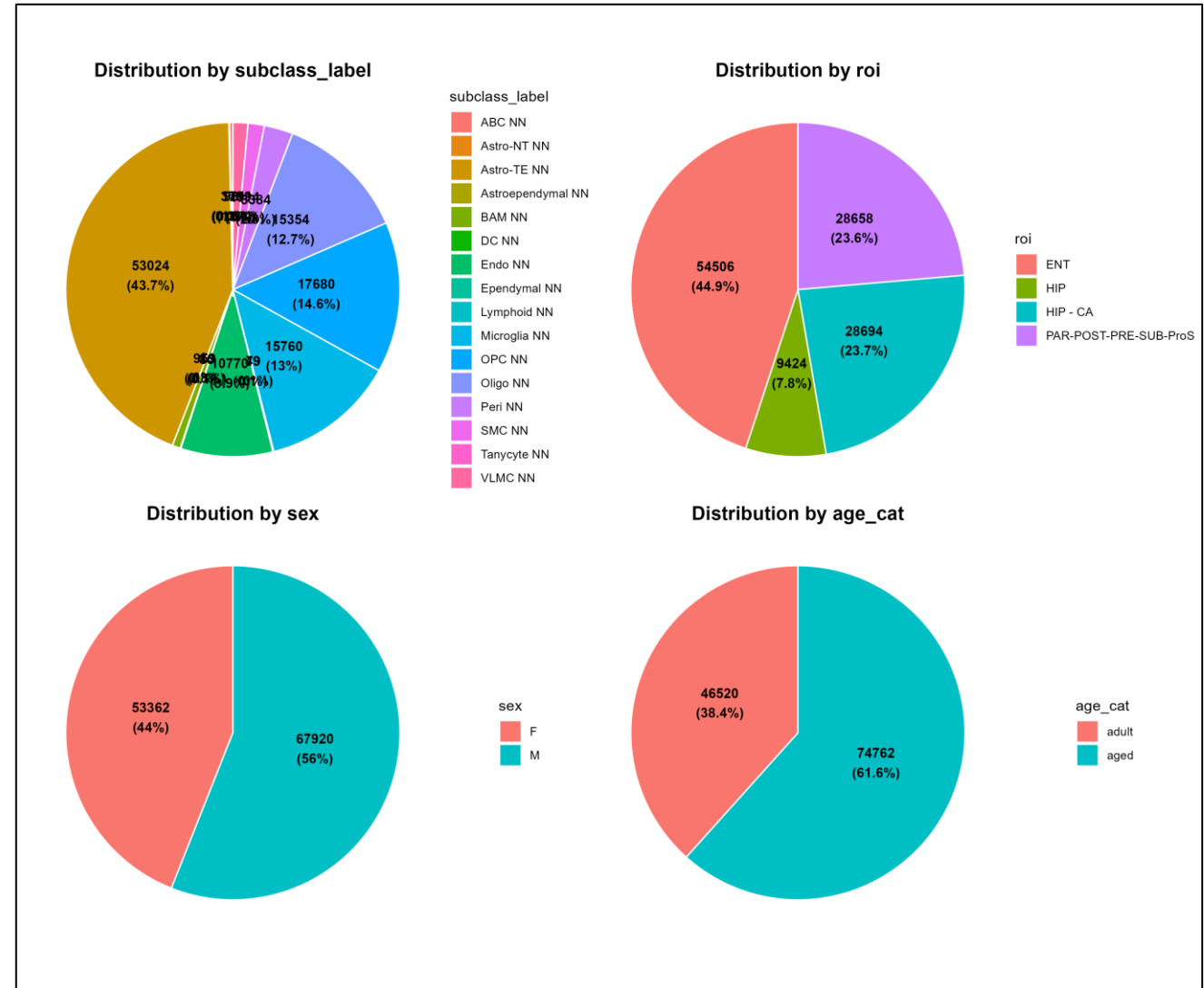
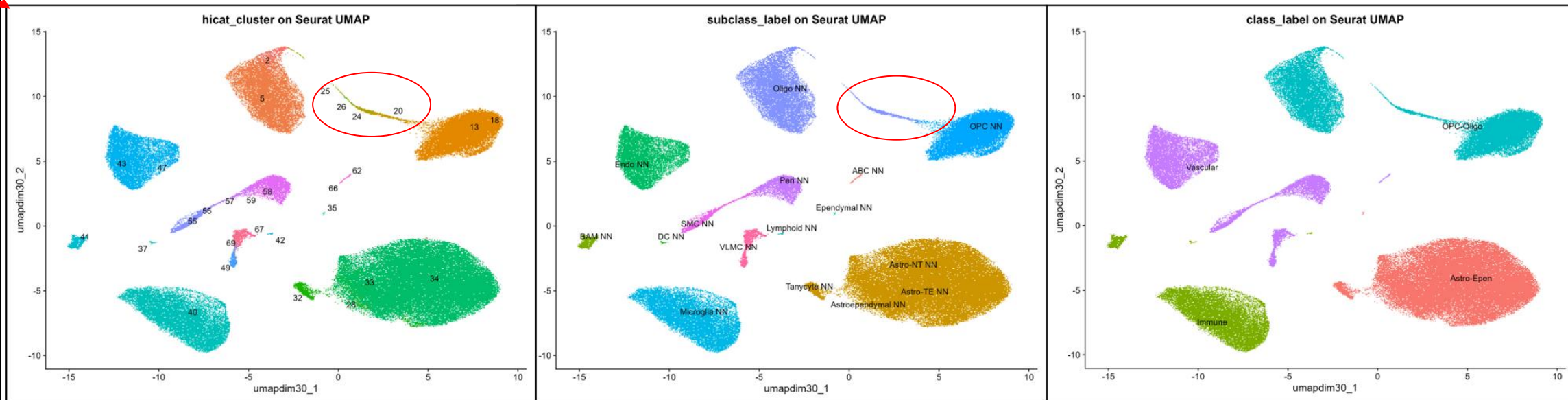


Figure 6: Cell demographic distribution (Created using ggplot2 R package; <https://ggplot2.tidyverse.org/>).

Annotation



Metadata

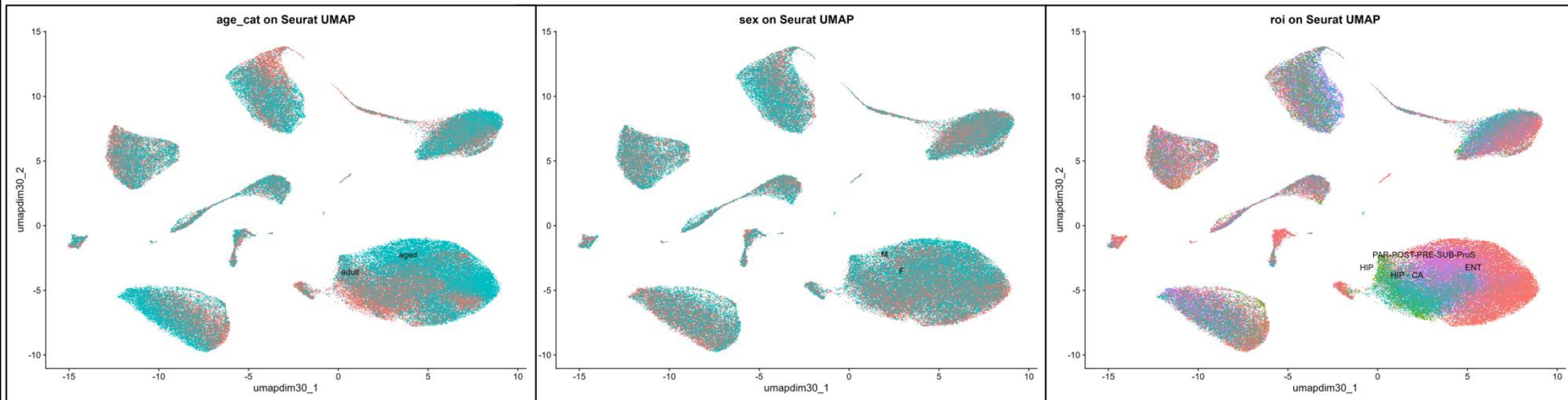


Figure 7: UMAP visualization grouped by annotation and metadata; The red arrow denote the clustering result; The red circle denote cluster identified from clustering but missed in sub_class (Created using Seurat R package; [Hao et al, 2023](#)).

Differential Gene Expression Analysis

Seurat ([Hao et al, 2023](#))

- FindAllMarkers()
 - test.use = wilcox
 - logfc.threshold = 0.1
 - min.pct = 0.01
 - return.thresh = 0.01

Scope (Analyzed Population)	Sex Category	Age Category	ROI Category ^[a]
Global (All subclass)	Male vs Female	Adult vs Aged	HPF Sub-region Comparisons
Local Subclass 1	Male vs Female	Adult vs Aged	Sub-region Comparisons
Local Subclass 2	Male vs Female	Adult vs Aged	Sub-region Comparisons
Local Subclass 3	Male vs Female	Adult vs Aged	Sub-region Comparisons
... (Iterated for all n=16 unique subclasses)	...	...	...

Table 1: Schematic of multi-axial Differential Expression (DE) comparison matrix; The table summarizes the systematic methodology employed to identify differentially expressed genes across glial cells of the hippocampal formation (HPF). ^[a] Regional (ROI) comparisons included contrasts between four major HPF anatomical structures, Entorhinal cortex (ENT), Hippocampus (HIP), Cornu Ammonis (HIP-CA), and Hippocampal-subicular (PAR_POST_PRE_SUB_ProS).

Global

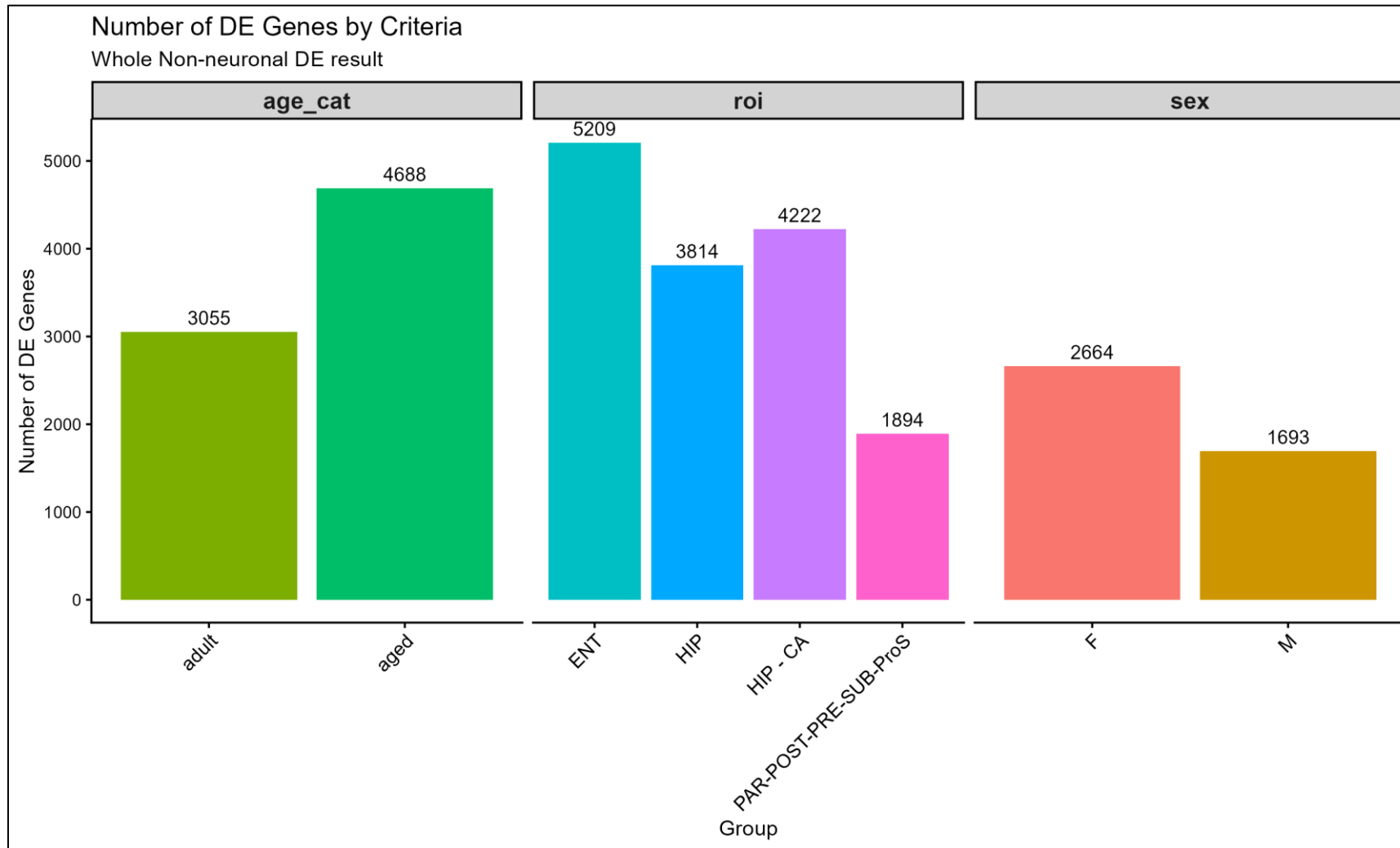


Figure 8: Global differentially expressed gene count; Y-axis is the number of significantly differentially expressed gene; X-axis are groups separated by age, roi, and sex category (Created using ggplot2 R package; <https://ggplot2.tidyverse.org/>).

Local

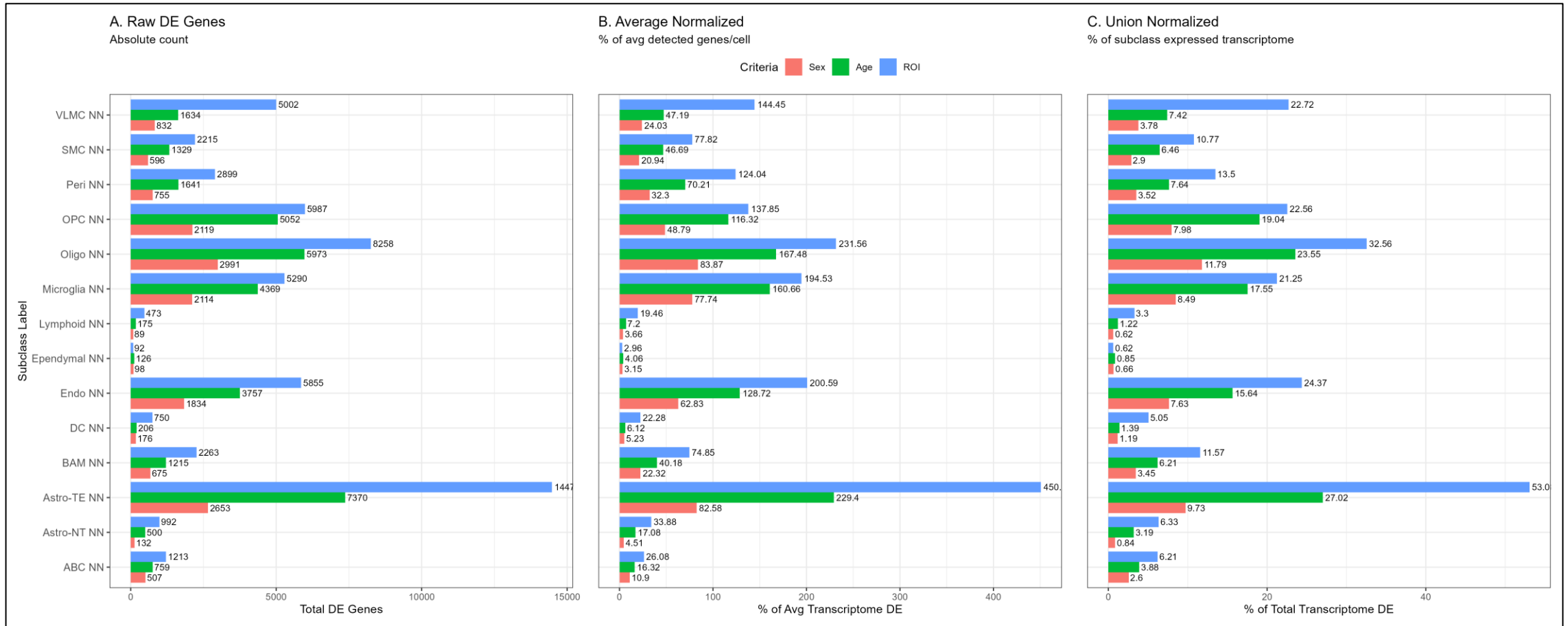


Figure 9: Local differentially expressed gene count; Y-axis is each subclass grouped by three categories, 14 out of 16 subclass contains differentially expressed gene, astroependymal and tanycyte doesn't contain differentially expressed gene; X-axis is the number of significantly differentially expressed gene; (A) raw counts; (B) normalized by average number of genes detected per cell; (C) normalized by the total number of genes detected per subclass (Created using ggplot2 R package; <https://ggplot2.tidyverse.org/>).

Gm47283

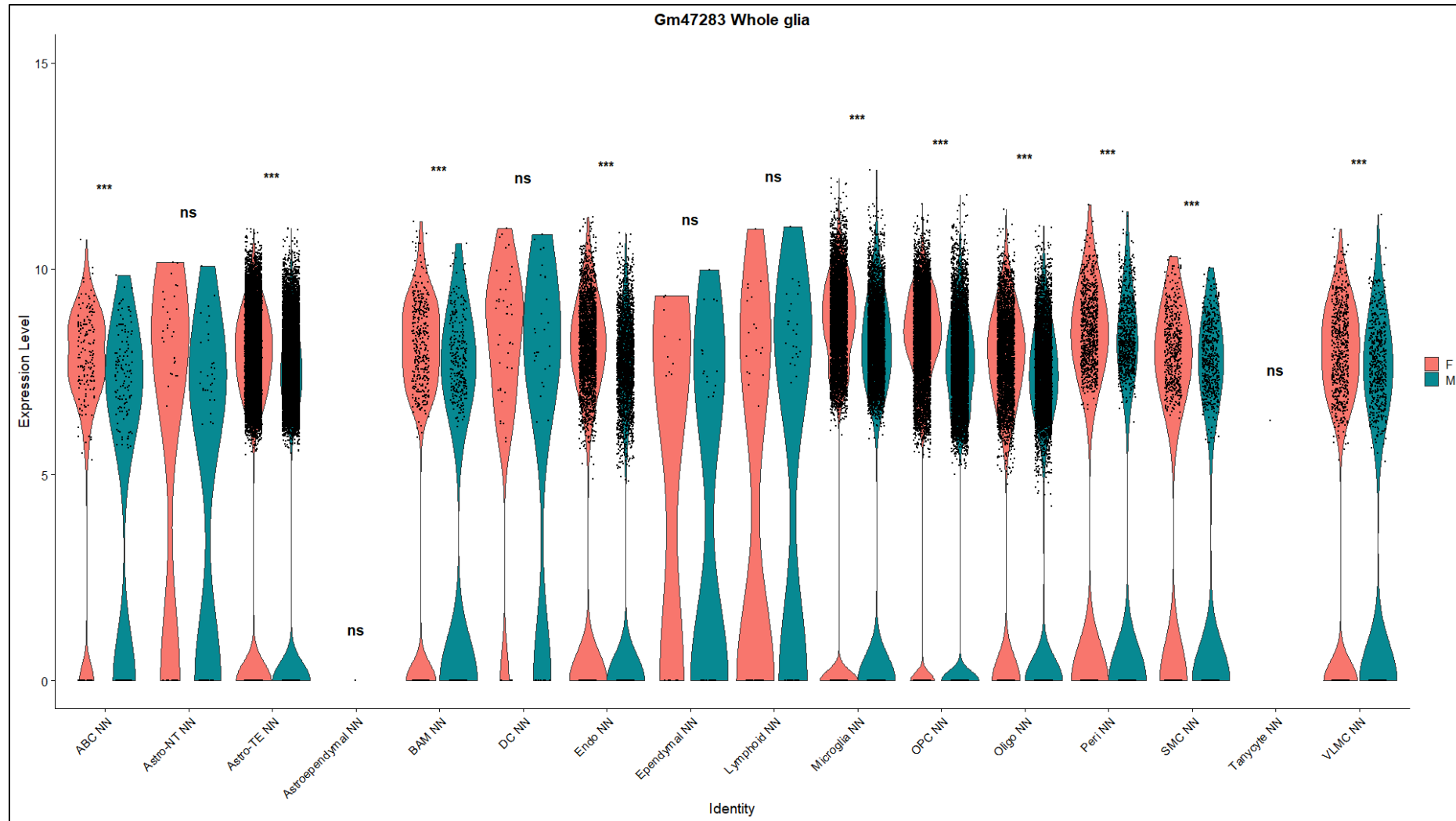


Figure 10: Expression level of *Gm47283* across 16 glial subclasses between sex, upregulated in female in 10 out of 16 subclass; Y-axis is the normalized expression level; X-axis all 16 glial subclass separated by sex; Significance levels are denoted above each subclass (***: adjusted $p < 0.001$; ns: not significant) (Created using Seurat R package; [Hao et al, 2023](#)).¹³

Synteny Exploration

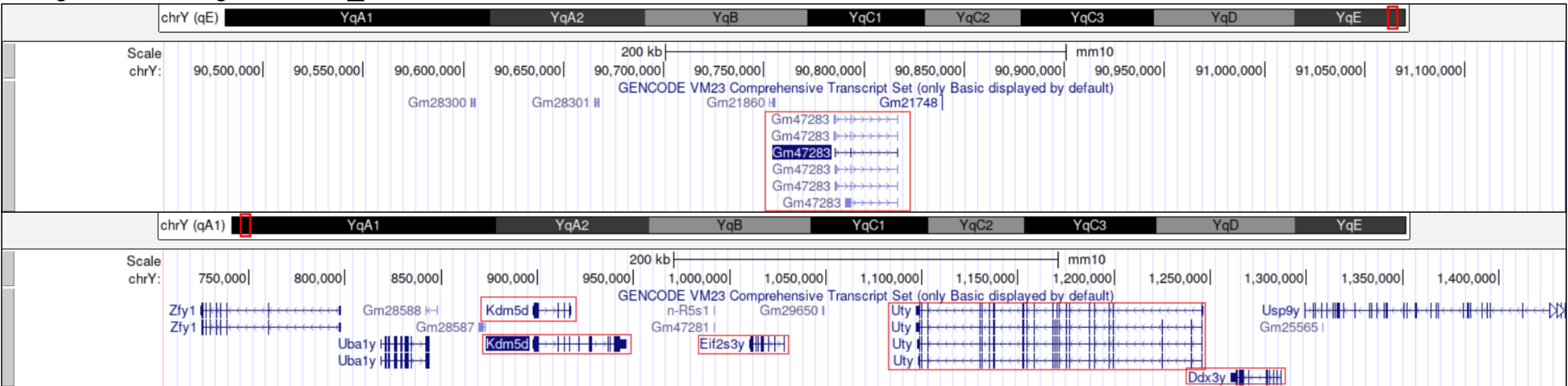


Figure 11: Synteny of top 5 Y-linked differentially expressed gene (*Gm47283*, *Kdm5d*, *Eif2s3y*, *Uty*, *Ddx3y*) on GRCm38/mm10 reference genome; Top panel indicate the synteny of *Gm47283*; Bottom panel indicate the synteny of *Kdm5d*, *Eif2s3y*, *Uty*, and *Ddx3y*; The red box on top of each panel indicate location on Y chromosome (Generated using UCSC genome browser; [Casper et al, 2026](#)).

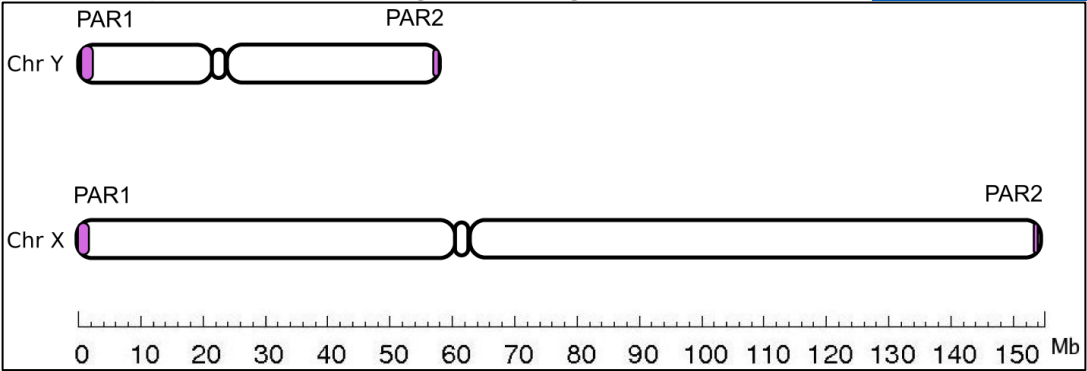


Figure 12: Schematic of human pseudoautosomal region highlighted in purple on each tip of sex chromosome.

Pseudoautosomal region

The origin and evolution of the pseudoautosomal regions of human sex chromosomes

J A Graves ¹, M J Wakefield, R Toder

Affiliations + expand

PMID: 9817914 DOI: [10.1093/hmg/7.13.1991](https://doi.org/10.1093/hmg/7.13.1991)

Abstract

The human X and Y chromosomes share two homologous pseudoautosomal regions (PARs) which pair and recombine at meiosis. PAR1 lies at the tips of the short arms, and the smaller PAR2 at the tips of the long arms. PAR1 contains several active genes, and has been thought to be critical for pairing and fertility. The inconsistent gene content of the PARs between different species of eutherian ('placental') mammals suggests that gene content is immaterial to function, and the failure to detect a PAR at all in some rodents and all marsupials implies that homologous pairing is not universally essential for fertility. The autosomal localization of marsupial homologues of human PAR1 genes and their co-localization with human Xp22 genes implies that the human PAR1 represents a relic of part of an autosomal region added to both X and Y chromosomes between 80 and 130 MYrBP. The same argument may be made for part of PAR2. Independent additions to the sex chromosomes of macropodid marsupials and monotremes can also be inferred from comparative mapping. We conclude that the PARs are relics of differential additions, loss, rearrangement and degradation of the Y chromosome in different mammalian lineages.

Figure 13: Abstract of Graves & Toder, 1998 ([Graves & Toder, 1998](#))

Reciprocal BLAST

- Pseudoautosomal region genes are homologous ([Graves & Toder, 1998](#); [Mange & Morris, 2007](#))

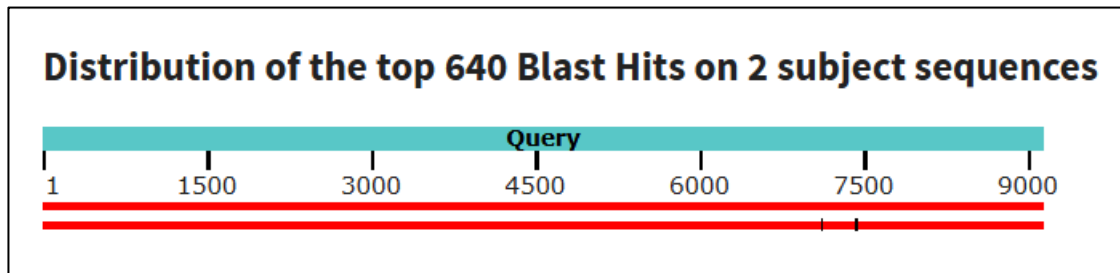
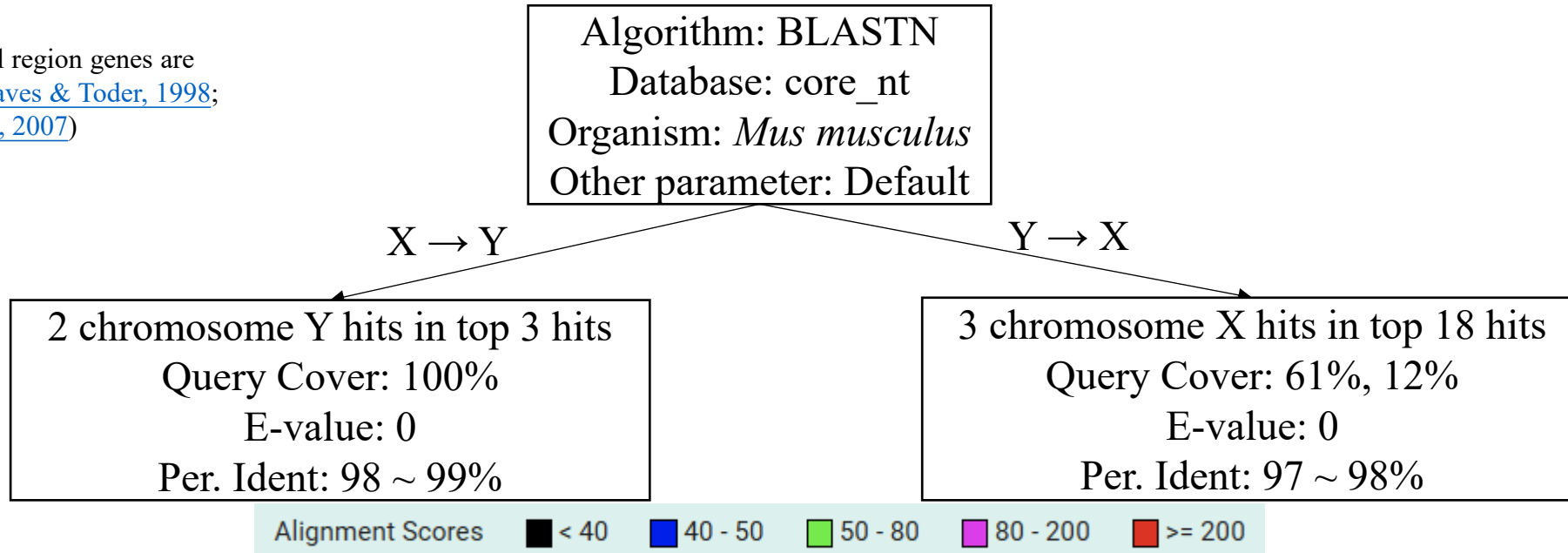


Figure 14: Graphical representation of *Gm47283* X to Y allele alignment (NCBI BLAST; [Sayers et al, 2025](#)).

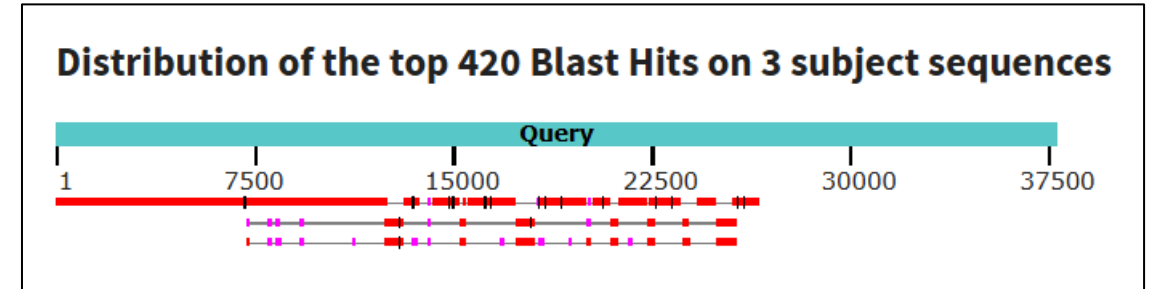


Figure 15: Graphical representation of *Gm47283* Y to X allele alignment (NCBI BLAST; [Sayers et al, 2025](#)).

Reads Mapping Algorithm Assessment

Jin et al, 2025 (<https://doi.org/10.1038/s41586-024-08350-8>)

Sequencing data preprocessing

All libraries were 10xv3 samples and processed as previously described^{3,20}. All libraries were sequenced on Illumina NovaSeq6000 and sequencing reads were aligned to the mouse reference (mm10/gencode.vM23)⁸⁸ using the 10x Genomics Cell Ranger pipeline (v.6.0.0) with the `-include introns` argument to include intronically mapped reads. This resulted in an scRNA-seq dataset containing 2,777,165 cells.

10x Genomics Cell Ranger official algorithm overview document (<https://www.10xgenomics.com/support/cn/software/cell-ranger/latest/algorithms-overview/cr-gex-algorithm>)

Genome alignment

Cell Ranger employs the **STAR** aligner for splicing-aware alignment of reads to the genome. It categorizes reads into exonic, intronic, or intergenic based on their alignment, using the transcript annotation GTF file. A read is classified as exonic if at least 50% of it overlaps with an exon. It is deemed intronic if it does not qualify as exonic but intersects an intron, and it is labeled intergenic if it fits neither of the previous categories.

Dobin et al, 2013
(<https://doi.org/10.1093/bioinformatics/bts635>)

2.2 Clustering, stitching and scoring

In the second phase of the algorithm, STAR builds alignments of the entire read sequence by stitching together all the seeds that were aligned to the genome in the first phase. First, the seeds are clustered together by proximity to a selected set of 'anchor' seeds. We found that an optimal procedure for anchor selection is through limiting the number of genomic loci the anchors align to. All the seeds that map within user-defined genomic windows around the anchors are stitched together **assuming a local linear transcription model**. The size of the genomic windows determines the maximum intron size for the spliced alignments. A frugal dynamic programming algorithm (see [Supplementary Section 1.5](#) for details) is used to stitch each pair of seeds, allowing for any number of mismatches but only one insertion or deletion (gap).

Hypothesis

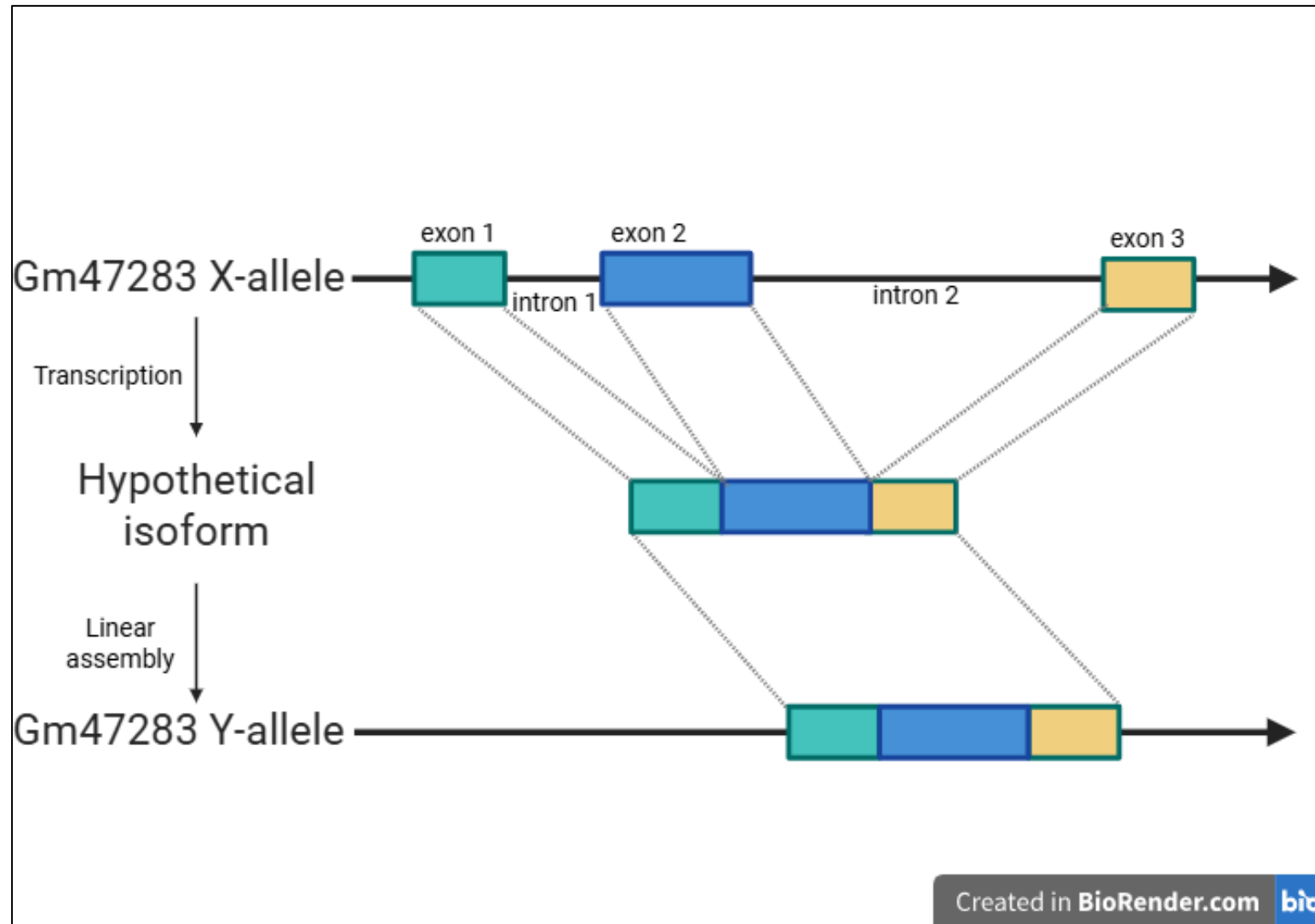


Figure 16: Proposed mechanism of *Gm47283* reads mismatching; The schematic illustrates a hypothetical algorithmic error during the reads mapping phase of *Gm47283*. The true genomic locus of the *Gm47283* X-allele (top) consists of defined exonic regions (boxes) separated by intronic sequences (lines). Transcription and splicing produce a mature isoform devoid of introns (middle). The hypothesis proposes that this processed, continuous isoform was mapped to continuous Y-allele (bottom) under linear assumption of mapping algorithm, resulting in a mismatching (created in <https://BioRender.com>).

Discussion

Hypothesis Validation

Dry

- X, Y allele pairwise alignment
- Isoform, alleles pairwise alignment
- Re-map female read to X-only reference genome

Wet

- Fluorescence In Situ Hybridization (FISH)

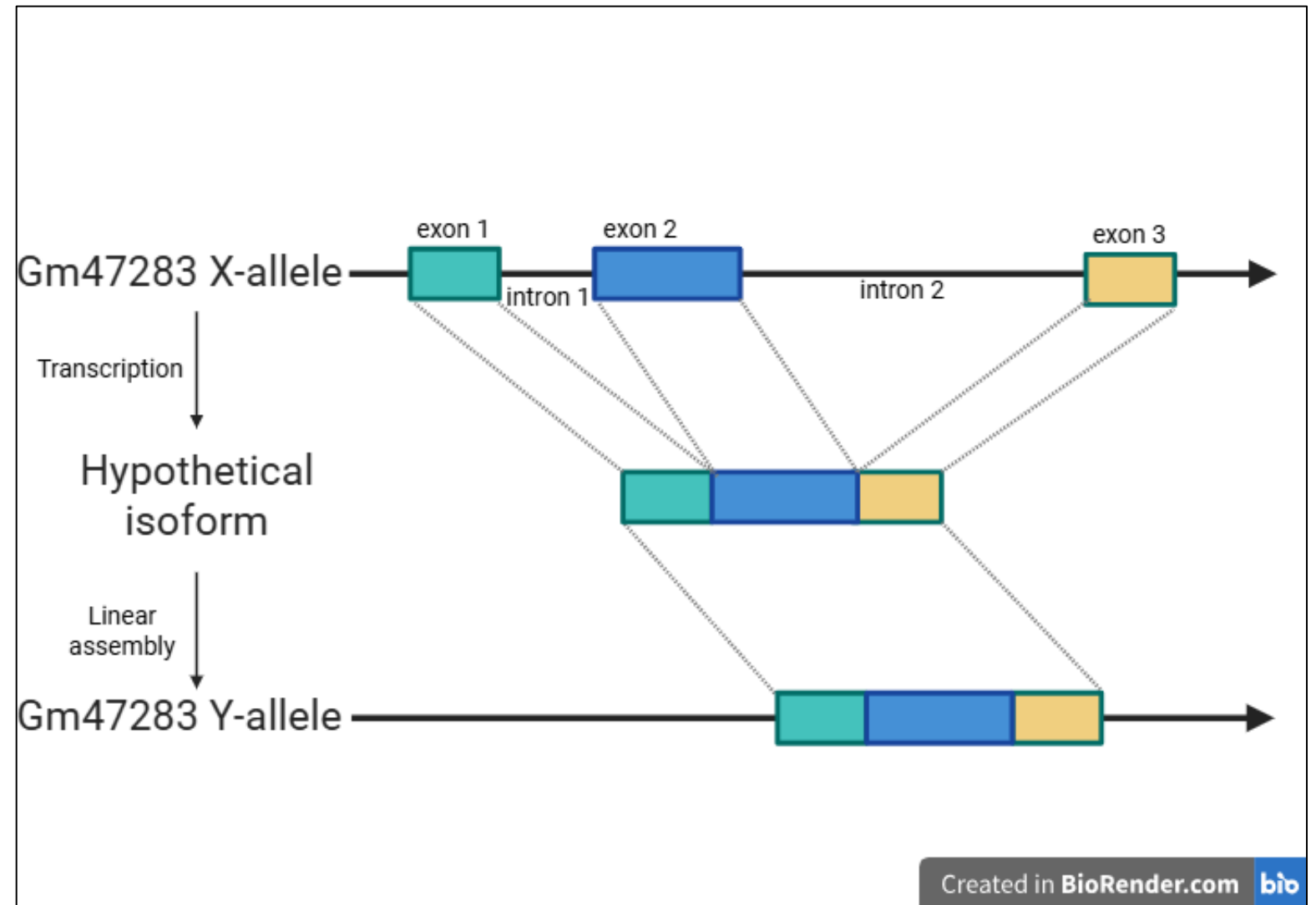


Figure 16: Proposed mechanism of *Gm47283* reads mismatching (created in <https://BioRender.com>).

Pseudoautosomal region

- scRNA-seq is not a suitable tool to study pseudoautosomal region.
- Is there any other genome region similar to pseudoautosomal region?
- Is there anything special about the hypothetical isoform compare with other isoform(s)?

Cell Type Deconvolution

- Problematic count matrix → Problematic cell type annotation
- False positive/negative sex-specific marker
- Significance of sex as biological variable

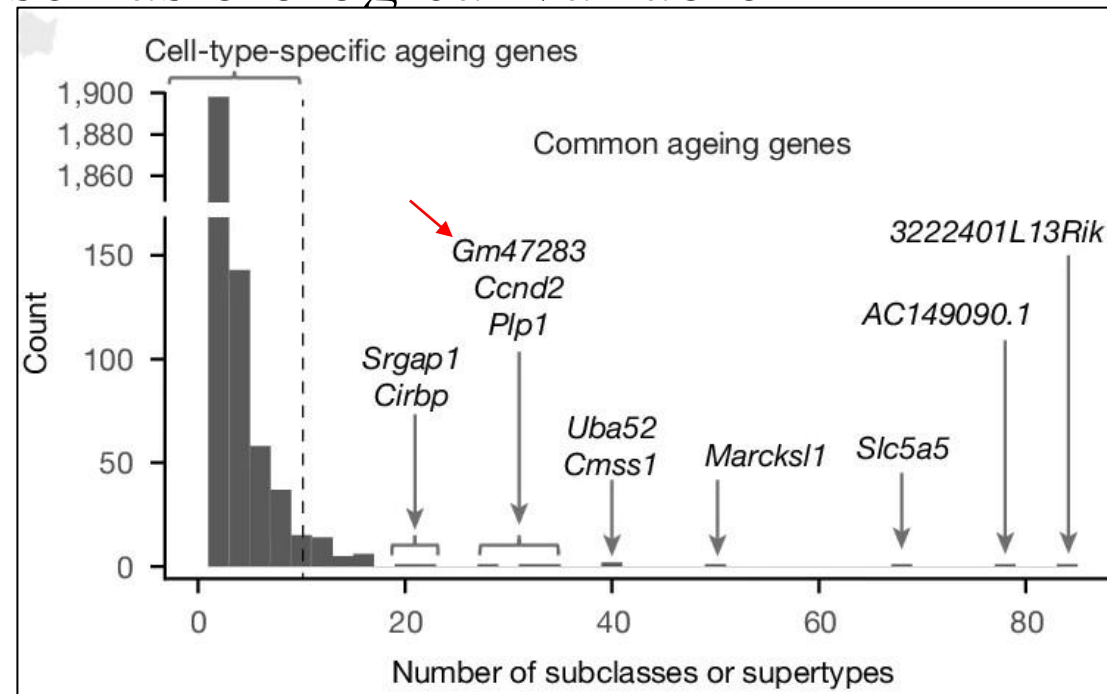


Figure 17: Histogram of number of significant DE genes among subclasses and supertypes; Y-axis is the number of DE genes; X-axis is subclasses or supertypes ([Jin et al, 2025](#), figure 1h).

Reference

- Pivotal
 - Jin, Kelly, Zizhen Yao, Cindy T. J. van Velthoven, Eitan S. Kaplan, Katie Glattfelder, Samuel T. Barlow, Gabriella Boyer, et al. 2025. “Brain-Wide Cell-Type-Specific Transcriptomic Signatures of Healthy Ageing in Mice.” *Nature* 638 (8049): 182–96. <https://doi.org/10.1038/s41586-024-08350-8>.
 - Hao, Yuhao, Tim Stuart, Madeline H. Kowalski, Saket Choudhary, Paul Hoffman, Austin Hartman, Avi Srivastava, et al. 2024. “Dictionary Learning for Integrative, Multimodal and Scalable Single-Cell Analysis.” *Nature Biotechnology* 42 (2): 293–304. <https://doi.org/10.1038/s41587-023-01767-y>.
 - “AllenInstitute/Scratch.hicat.” 2026. Allen Institute. <https://github.com/AllenInstitute/scratch.hicat>.
 - Casper, Jonathan, Matthew L. Speir, Brian J. Raney, Gerardo Perez, Luis R. Nassar, Christopher M. Lee, Angie S. Hinrichs, et al. 2026. “The UCSC Genome Browser Database: 2026 Update.” *Nucleic Acids Research* 54 (January): D1331–35. <https://doi.org/10.1093/nar/gkaf1250>.
 - “Genome Reference Consortium.” n.d. Accessed March 27, 2026. <https://www.ncbi.nlm.nih.gov/grc>.
 - Sayers, Eric W., Jeffrey Beck, Evan E. Bolton, J. Rodney Brister, Jessica Chan, Ryan Connor, Michael Feldgarden, et al. 2025. “Database Resources of the National Center for Biotechnology Information in 2025.” *Nucleic Acids Research* 53 (January): D20–29. <https://doi.org/10.1093/nar/gkae979>.
 - “Cell Ranger’s Gene Expression Algorithm Official 10x Genomics Support. 10x Genomics.” n.d. Accessed March 27, 2026. <https://www.10xgenomics.com/support/software/cell-ranger/latest/algorithms-overview/cr-gex-algorithm>.
 - Dobin, Alexander, Carrie A. Davis, Felix Schlesinger, Jorg Drenkow, Chris Zaleski, Sonali Jha, Philippe Batut, Mark Chaisson, and Thomas R. Gingeras. 2013. “STAR: Ultrafast Universal RNA-Seq Aligner.” *Bioinformatics* 29 (1): 15–21. <https://doi.org/10.1093/bioinformatics/bts635>.
- Exhaustive reference list
 - <https://github.com/DDMMMAA/scratch.hicat-BCB430/blob/master/report/BCB430.bib>